

MCQ-10M

1.What is the default return value of a function that doesn't have a return statement?

- a) None
- b) 0
- c) False
- d) Null

2.What is the correct syntax for a function definition in Python?

- a) def functionName[]:
- b) define functionName():
- c) def functionName():
- d) function functionName():

3.Which of the following is used to exit a function and return a value?

- a) break
- b) continue
- c) exit
- d) return

4.Which of these is a correct example of a function with default parameters?

- a) def test(a, b=5):
- b) def test(a=5, b):
- c) def test(a+b=5):
- d) def test(a: 5, b):

5.Which keyword is used to allow a function to accept an arbitrary number of arguments?

- a) *
- b) varargs

c) **

d) args

6. In Python, function arguments are:

a) Pass by reference

b) Pass by value

c) Pass by object reference

d) Pass by pointer

7. What will `print(type(lambda x: x+1))` output?

a) `<class 'function'>`

b) `<function>`

c) `<lambda>`

d) `<class 'lambda'>`

8. Which function returns the number of arguments passed to a function?

a) `len()`

b) `args()`

c) `count()`

d) None of the above

9. What is the output of:

```
def f(x=[]):  
    x.append(1)  
    return x  
  
print(f())  
print(f())
```

a) `[1] [1]`

- b) [1, 1] [1, 1]
- c) [1] [1, 1]
- d) Error

10. What type of function is map() in Python?

- a) Recursive
- b) Lambda
- c) Built-in
- d) User-defined

Section B: 2-Mark

5 × 2 = 10 M

1. Differentiate between *args and **kwargs.
2. What are recursive functions? Give one example.
3. What is a lambda function? Write its syntax.
4. Explain the concept of scope with respect to functions.
5. What is the difference between a function definition and a function call?

Section C: Coding

6 × 5 = 30 M

1. Write a Python function to check whether a number is prime or not.
2. Write a function count_vowels() that takes a string and returns the number of vowels in it.
3. Write a recursive function to calculate the factorial of a number.
4. Create a function is_palindrome() that checks whether a given string is a palindrome.
5. Write a Python function that accepts a variable number of arguments and returns their sum.
6. Define a function max_of_three(a, b, c) that returns the maximum of three numbers without using the built-in max() function.